Quartus Prime Lite Edition

Home Description Diagrams Pictures Bugs Future Work-Distribution

ECE241 FINAL PROJECT

PACMAN

BY: IREEN & MARIAM

Start

Quartus Prime Lite Edition

DEFINING PACMAN

- made our own version of the original 1980s Pac-man arcade game



[1] C. Boltan, Pacman Games Through Time (Image). 2023.

Quartus Prime Lite Edition

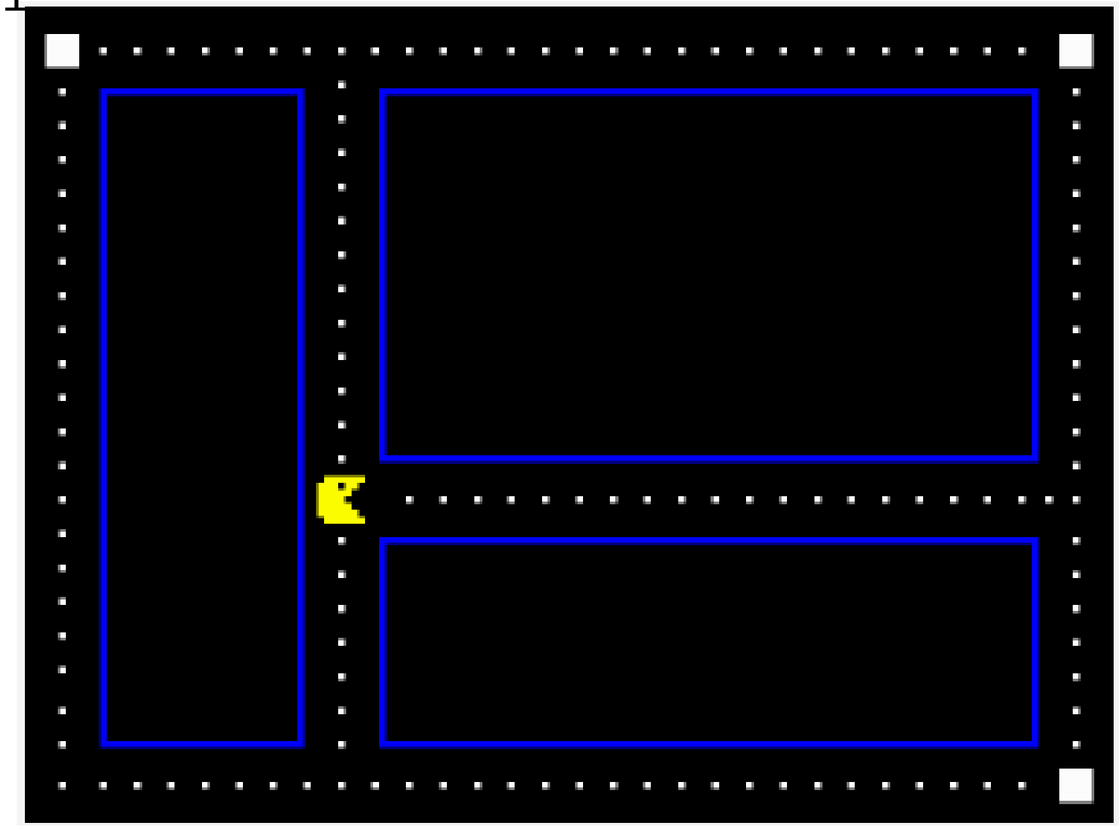
- features **one** ghost, **three** coins, multiple mini coins, & **Pac-man!**

- user can move in **all directions** using **WASD**
- Pac-man can move past the edge of the screen and come back the other side

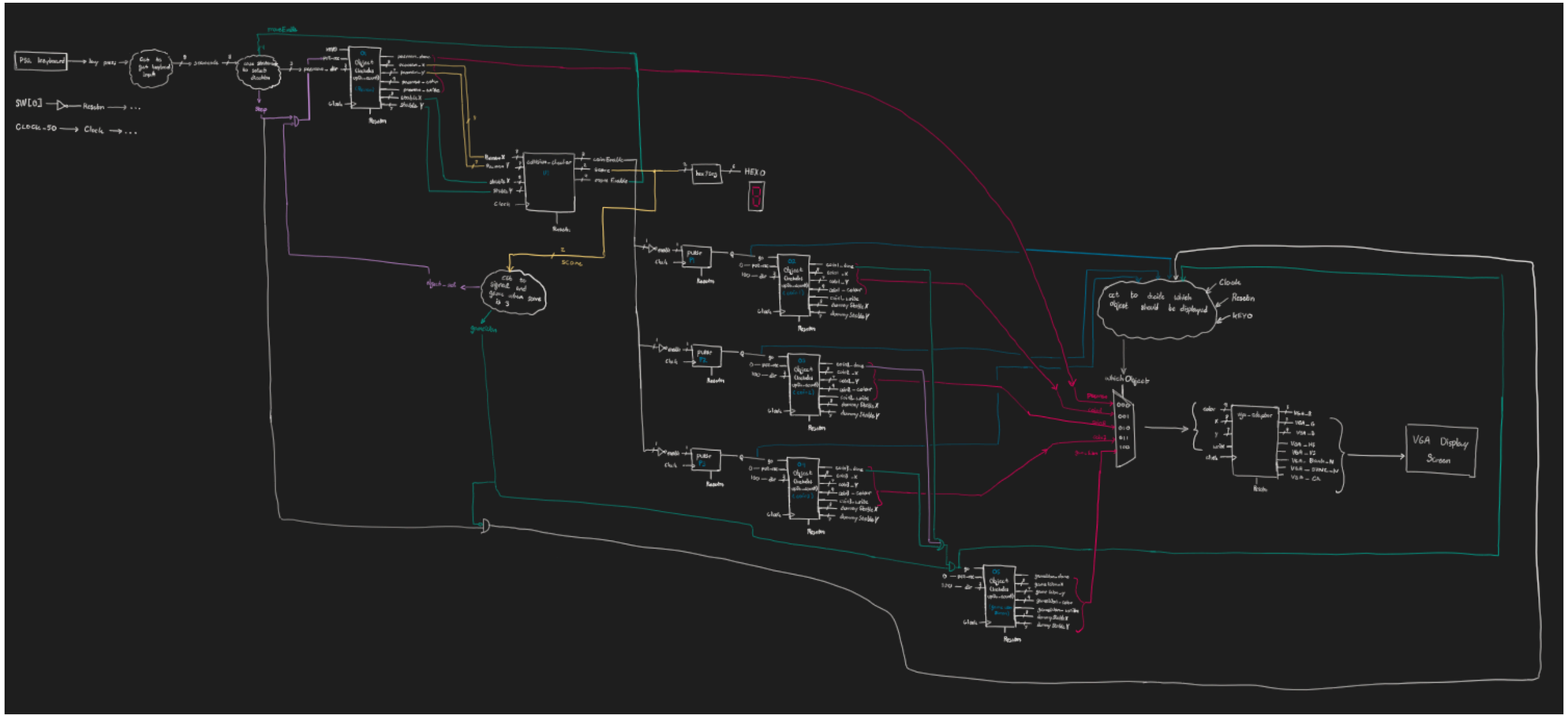
- the score increments every time there is a collision with a big coin
- mini-coins do not give any points
- a score of three means **GAME WON!**

- maze consists of three bounding boxes

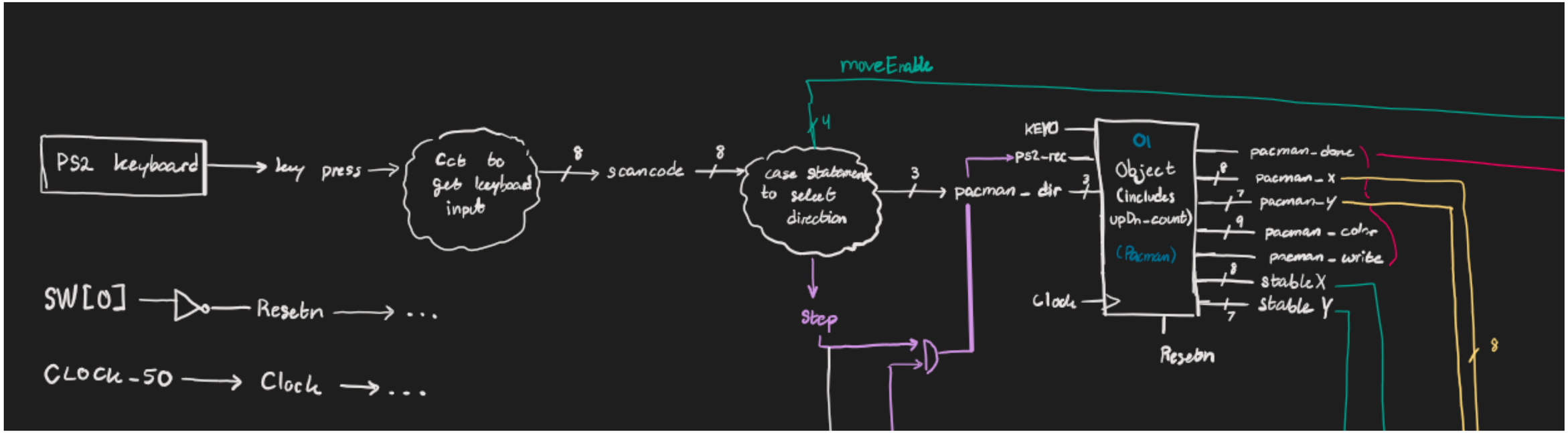
- **extra =>** ghost traverses a set path
- collision with the ghost is **GAME OVER!**

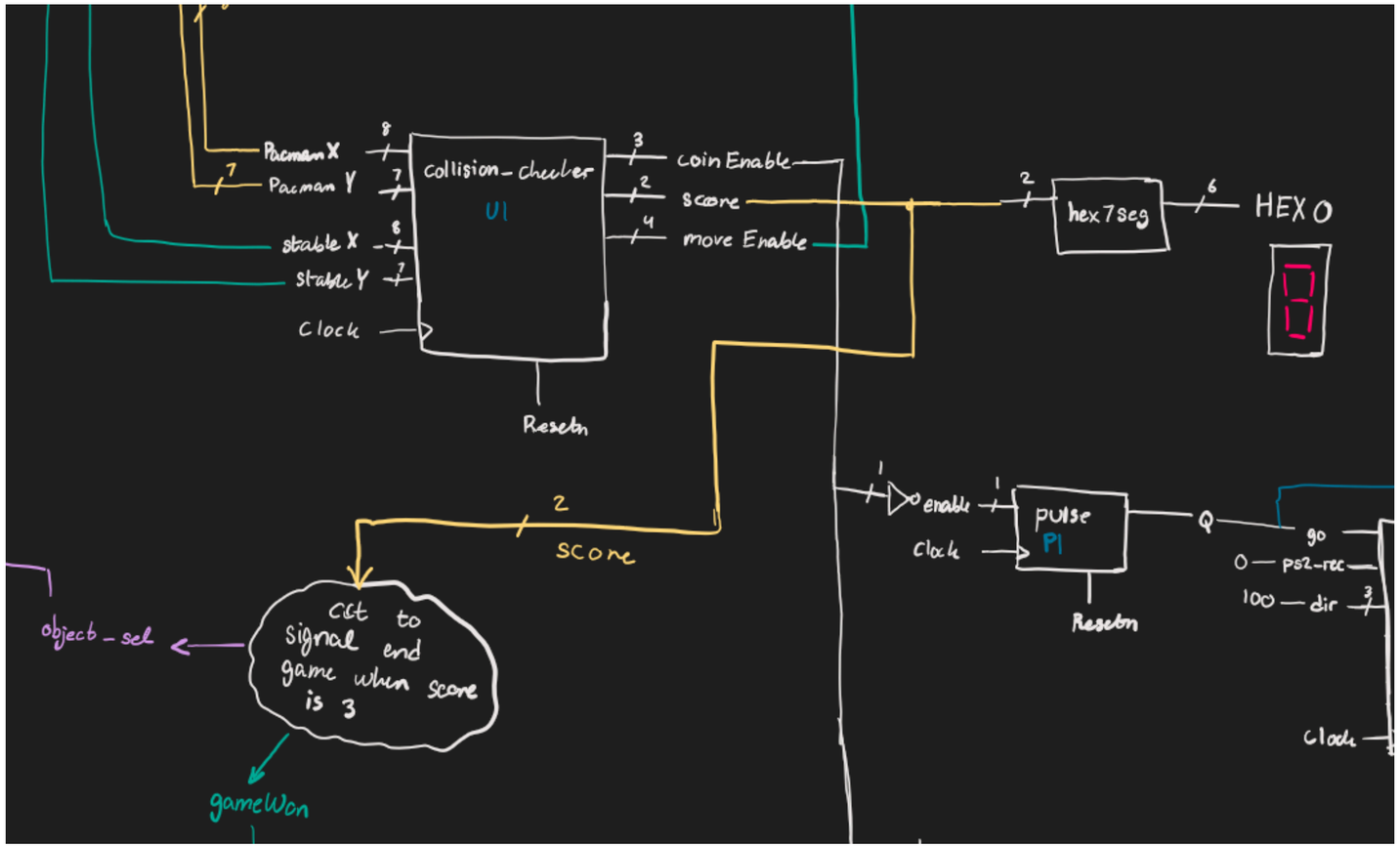


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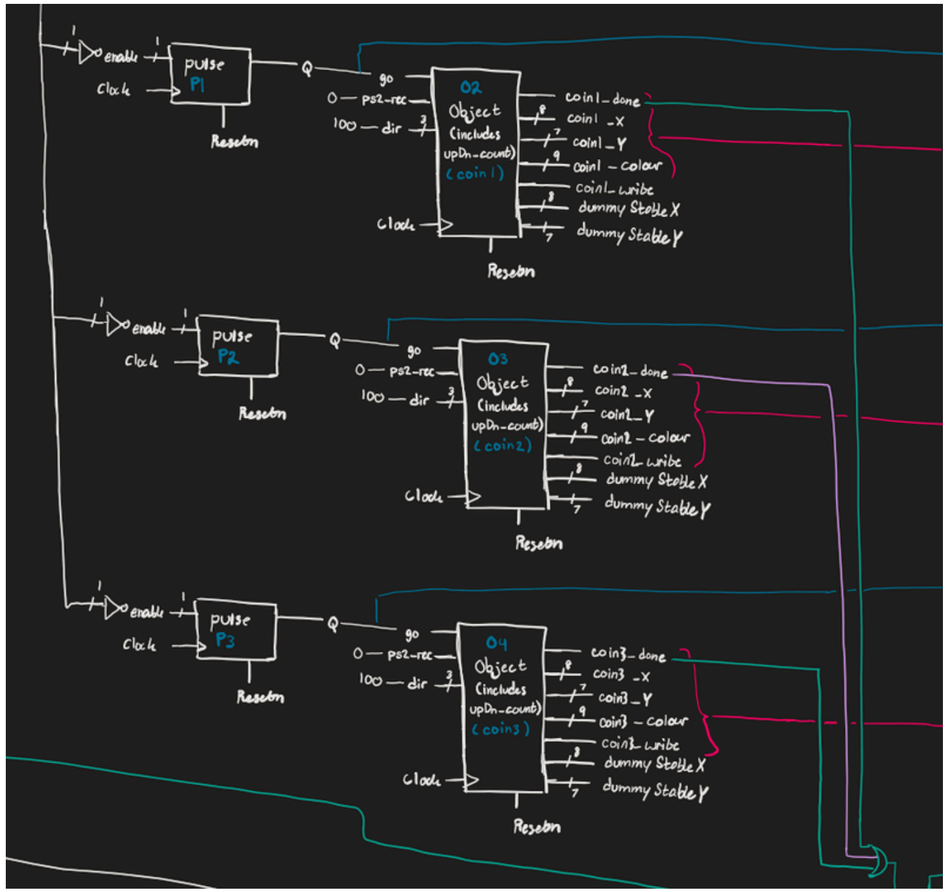


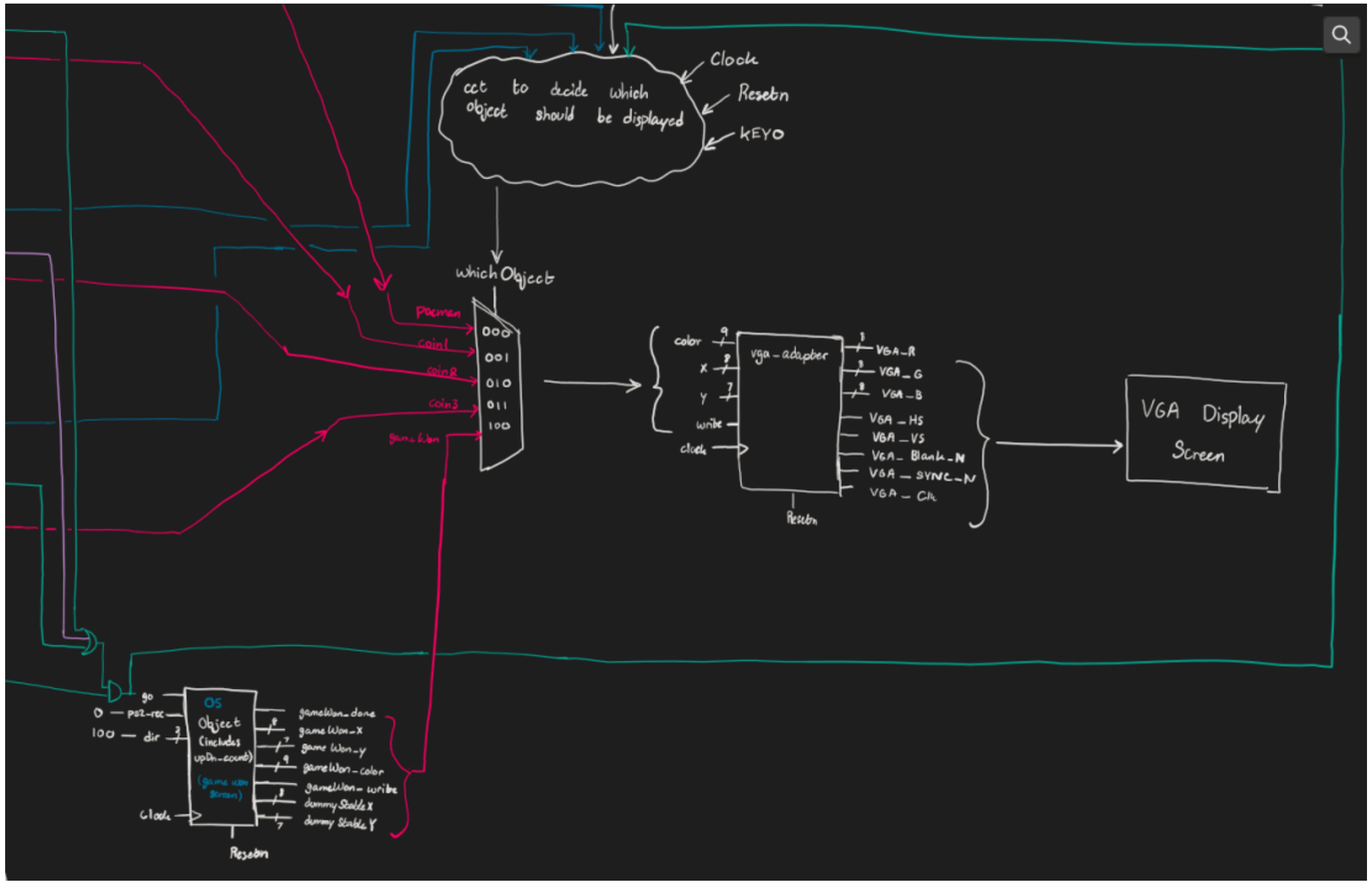
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Pacman X
Pacman Y
stable X
stable Y
Clock
collision_checker
U1
Resetn
coin Enable
score
move Enable
hex7seg
HEX 0
2
score
enable
Clock
pulse
P1
Resetn
Q
go
0 — ps2_rec
100 — dir
Clock
object_sel
Set to signal end game when score is 3
gameWon

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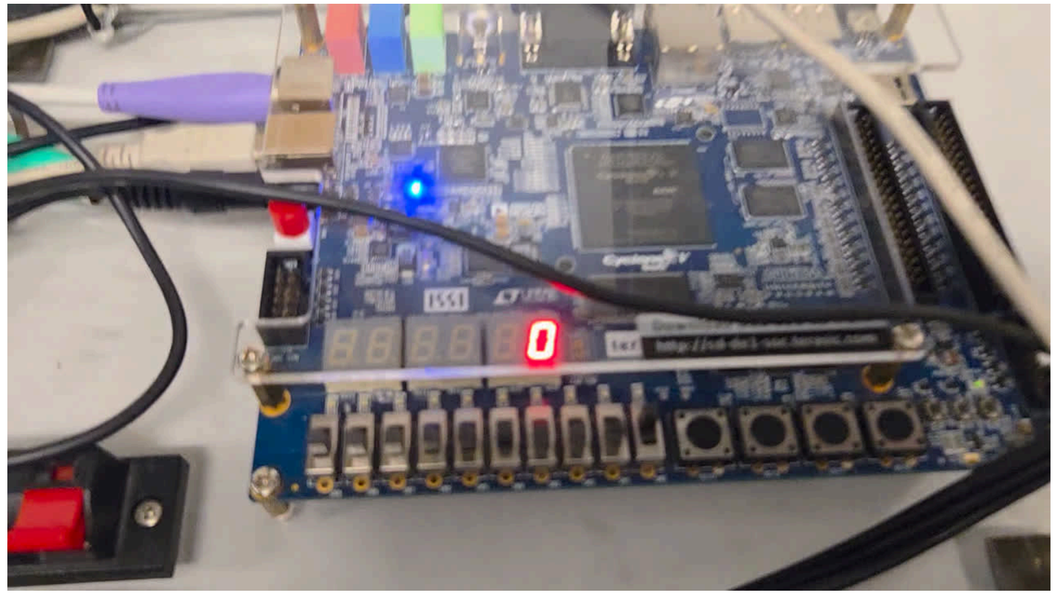


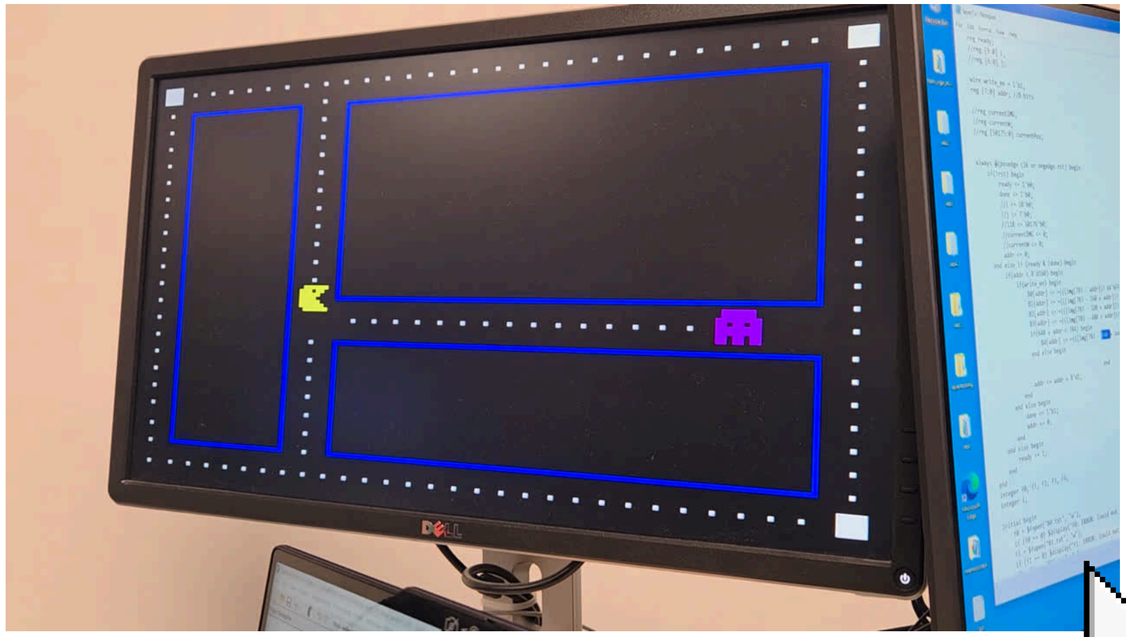


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VIDEOS FOR PROOF!





^^ semi-working game w/Pac-man, ghost & coins

<= working game with only Pac-man & coins

BUGS & ISSUES

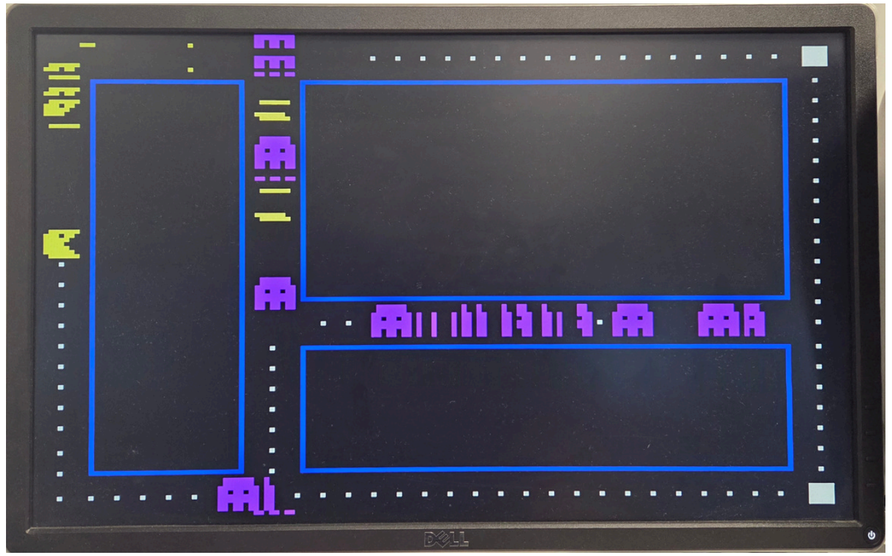
- getting the keyboard to only use **WASD** and no other key for movement

- maze collisions would break => Pac-man would **collide with walls that were not there**
- Pac-man’s **movement would never be disabled**

- coins would not disappear instantly after collisions
- **score would immediately increment to one** after reset and the first coin would be disabled

- **game won screen would not display** when third coin disappeared or vice versa

- **ghost would not change direction** when necessary
- ghost would spawn in the wrong location



HOW WE FIXED IT!

- **keyboard =>** was a port size error, created a new “don’t move” direction
- **maze collisions =>** another port size error, maze coordinates were off by 7 pixels
- **coin collisions =>** created a pulse module that would emulate a key press, fixed coin collision logic
- **game win screen =>** only drew GAME WON! after the third coin was done drawing
- **ghost path =>** changed the path we chose and offset the coordinates



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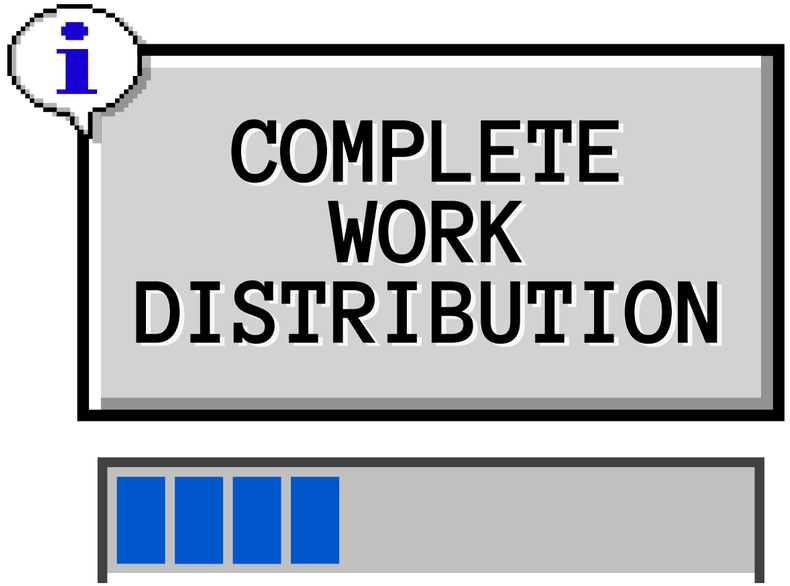


- properly implement ghost + Pac-man drawing with no bugs
- make screen redraw on reset
- add sounds to the game (e.g. Pac-man moving, coin collision)
- increment the score when the mini coins are eaten & draw score on screen
- implement a timer system to add another level of difficulty
- have more ghosts moving around
- improve path of the ghost => make it turn around and go over the coins but not delete them
- make Pac-man face in a direction based on the key that is being pressed
- add cherries to the game that increase the speed of Pac-man when eaten
- make the maze more complicated rather than having only 3 bounding boxes

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IREEN

- Made pixel art for pacman, coins, and the maze
- Modified demo code to draw moveable objects and a static background
- Wrote code to make Pacman’s movement scroll smoothly
- Implemented coin and maze collision logic with the game itself
- Made the coins disappear immediately when collided with
- Made the game won screen appear after all coins were collected
- Implemented keyboard so that Pacman only moves using WASD and no other keys
- Created presentation slides
- Wrote code to make Pacman appear at the start of the game, without having to press KEYs



MARIAM

- Wrote code for checking collision with coins and maze walls, including updating score and movement enable
- Created ModelSim testbenches to simulate code
- Wrote game logic for ghost movement
- Wrote game logic to check ghost collision with Pacman
- Made pixel art for ghost, game won screen and game over screen
- Displayed score on the hex display
- Created presentation slides
- Wrote code to allow Pacman and ghost to move at the same time
- Make game over screen appear when ghost collides with Pacman

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THANK YOU!

Made By

Mariam Neemuchwala

Ireen Cheralody

